2.7 Ribosomes and Mitochondria

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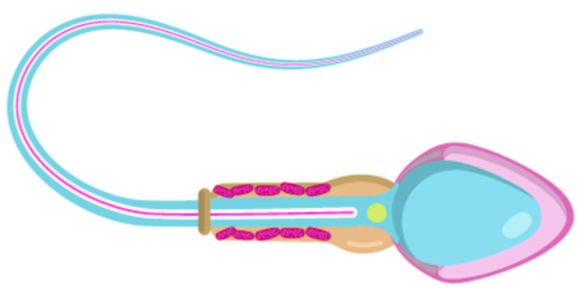
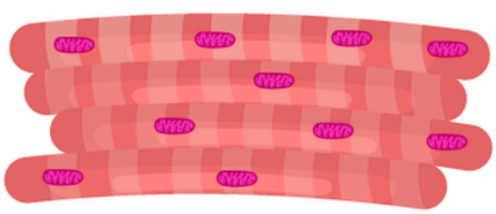
Last Modified: May 01, 2026

What you will learn

- The structure of **ribosomes** and their role in protein synthesis
- Why **mitochondria** are known as the "power plants" of the cell and how they meet the energy needs of the cell





[Figure 1]

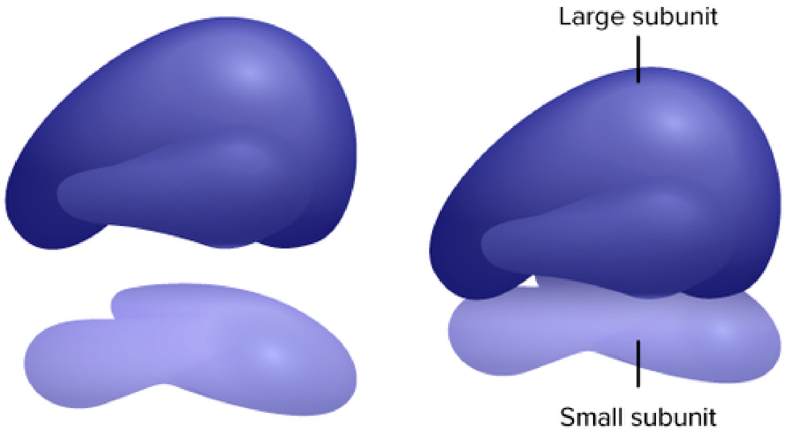
m cells and muscle cells need lots of energy. What do they have in common?

Ribosomes and Mitochondria

In addition to the nucleus, eukaryotic cells have many other cellular structures and organelles, including ribosomes and mitochondria. Ribosomes are cellular structures present in all cells.

Ribosomes

Ribosomes are small cellular structures that are the sites of protein synthesis (or assembly). They are made of ribosomal protein and ribosomal RNA, and are found in both eukaryotic and prokaryotic cells. Unlike organelles, ribosomes are not surrounded by a membrane. Each ribosome has two parts, a large and a small subunit, as shown in the **Figure** below. The subunits are attached to one another. Ribosomes can be found alone or in groups within the cytoplasm. Some ribosomes are attached to the endoplasmic reticulum (ER), and others are attached to the nuclear envelope.



[Figure 2]

The two subunits that make up a ribosome, small organelles that are intercellular protein factories.

Ribozymes are RNA molecules that catalyze chemical reactions, such as translation. **Translation** is the process of ordering the amino acids in the assembly of a protein, and translation will be discussed more in another concept. Briefly, the ribosomes interact with other RNA molecules to make chains of amino acids called polypeptide chains, due to the peptide bond that forms between individual amino acids. Polypeptide chains are built from the genetic instructions held within a messenger RNA (mRNA) molecule. Polypeptide chains that are made on the rough ER are inserted directly into the ER and then are transported to their various cellular destinations. Ribosomes on the rough ER usually produce proteins that are destined for the cell membrane.

DID YOU KNOW?

Although ribosomes are generally described as organelles, it is important to note that ribosomes are not organelles, they are not bound by a membrane, and are much smaller than other organelles.

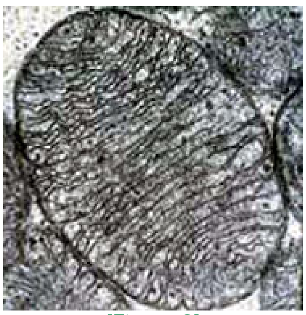
somes are found in both eukaryotic and prokaryotic cells. Ribosomes are not surrounded by a ıbrane. The other organelles found in eukaryotic cells are surrounded by a membrane.

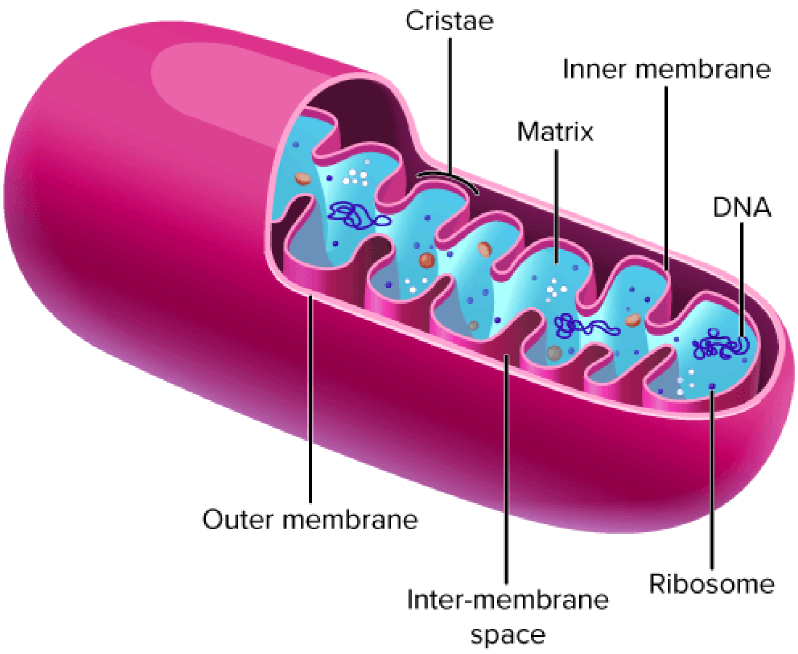
Translation (advanced)
Teacher's Pet

Watch on

Mitochondria

A mitochondrion (**mitochondria**, plural), is a membrane-enclosed organelle that is found in most eukaryotic cells. Mitochondria are called the "power plants" of the cell because they are the sites of cellular respiration, where they use energy from organic compounds to make **ATP** (adenosine triphosphate). **ATP** is the cell's energy source that is used for such things such as movement and cell division. Some ATP is made in the cytosol of the cell, but most of it is made inside mitochondria. The number of mitochondria in a cell depends on the cell's energy needs. For example, active human muscle cells may have thousands of mitochondria, while less active red blood cells do not have any.





- [Figure 3] [Figure 4]

Electron micrograph of a single mitochondrion, within which you can see many cristae. Mitochondria range from 1 to 10 µm in size. (second) This model of a mitochondrion shows the organized arrangement of the inner and outer

As the **Figure**above shows, a mitochondrion has two phospholipid membranes. The smooth outer membrane separates the mitochondrion from the cytosol. The inner membrane has many folds, called **cristae**. The fluid-filled inside of the mitochondrion, called the **matrix**, is where most of the cell's ATP is made.

DID YOU KNOW?

Prokaryotes lack mitochondria and instead produce ATP on their cell membrane.

Although most of a cell's DNA is contained in the cell nucleus, mitochondria have their own DNA. Mitochondria are able to reproduce asexually, and scientists think that they are descended from prokaryotes. According to the endosymbiotic theory, mitochondria were once free-living prokaryotes that infected or were engulfed by ancient eukaryotic cells. The invading prokaryotes were protected inside the eukaryotic host cell, and in turn the prokaryote supplied extra ATP to its host.

Summary

- Ribosomes are small cellular structures and are the site of protein synthesis. Ribosomes are found in all cells.
- Mitochondria are where energy from organic compounds is used to make ATP.

Review

1. What is the function of a ribosome?
2. What is a significant difference between the structure of a ribosome and other organelles?
3. Identify the reason why mitochondria are called "power plants" of the cell.
4. Describe the structure of a mitochondrion.